

Homework 6: Regression

Submission Notes: Please write your answers to each question under that question (not altogether at the end). It will be easier for me to give full credit if your answer is commented well, including comments about where you are completing each sub-part of the question. For an example, see Homework 1.

1. In Lecture 13, we discussed the linear algebra formulation of linear regression. For example, we can set up perfectly simulated regression data as follows:

```
simulate.reg <- function(n=100,
                        betas=c(4.35, 1.73, 0.02, 2.78),
                        sigma=4.16,
                        set.seed=F){
  # Specify Coefficients
  beta <- matrix(data = betas, nrow=4)
  # Specify Error Distribution
  mu.epsilon <- 0
  sigma.epsilon<-sqrt(sigma)
  # Simulate Regression Data
  if(set.seed){set.seed(7272)}
  x1 <- runif(n=n, min=-5, max=5)
  x2 <- runif(n=n, min=-5, max=5)
  x <- cbind(1, x1, x2, x1*x2)
  epsilon <- rnorm(n=n, mean=0, sd=sigma.epsilon)
  y <- x %*% beta + epsilon
  # Return simulated data
  dat <- tibble(y=y, x1=x1, x2=x2)
  return(dat)
}
```

In Lecture 15, we discussed regression-based inference. During office hours, I had some discussions about the mathematical statistics behind these results. Although we don't do the real-analysis style derivations in this course that we might in Mathematical Statistics, we have plenty of tools to investigate these ideas numerically. Complete the parts below to conduct this exploration.

- (a) Use the `simulate.reg()` function to simulate regression data using the default set of coefficients over a grid of σ_ϵ values from $\sigma_\epsilon = 0$ to $\sigma_\epsilon = 5$. Compute the SS_{res} and SS_{reg} from the resulting regression model, and plot the points using color to denote the σ_ϵ value. Make sure you set `set.seed` to T so you can compare the output at fixed **x** and **Y**. **Why do you see the pattern that emerges?**

Hint: I found it was easier to manually compute SS_{res} and SS_{reg} than it was to extract them from the result of an `aov` call.

```

asim <- tibble(sigma = rep(NA, 1000),
               SSres = rep(NA, 1000),
               SSreg = rep(NA, 1000))

sigmas <- seq(from = 0, to = 5, length.out = 1000)

for(i in 1:1000){

  dat <- simulate.reg(sigma = sigmas[i], set.seed=T)

  asim$sigma[i] <- sigmas[i]

  anova <- aov(y ~ x1 + x2 + x1*x2, data = dat)

  asim$SSres[i] <- sum((anova$residuals)^2)

  asim$SSreg[i] <- sum((predict(anova) - mean(dat$y))^2)

}

ggplot(data = asim, aes(x = SSres, y = SSreg, color = sigma)) +
  geom_point() +
  xlab(bquote(SS[res])) +
  ylab(bquote(SS[reg])) +
  labs(color = bquote(sigma[epsilon])) +
  theme_bw()

```

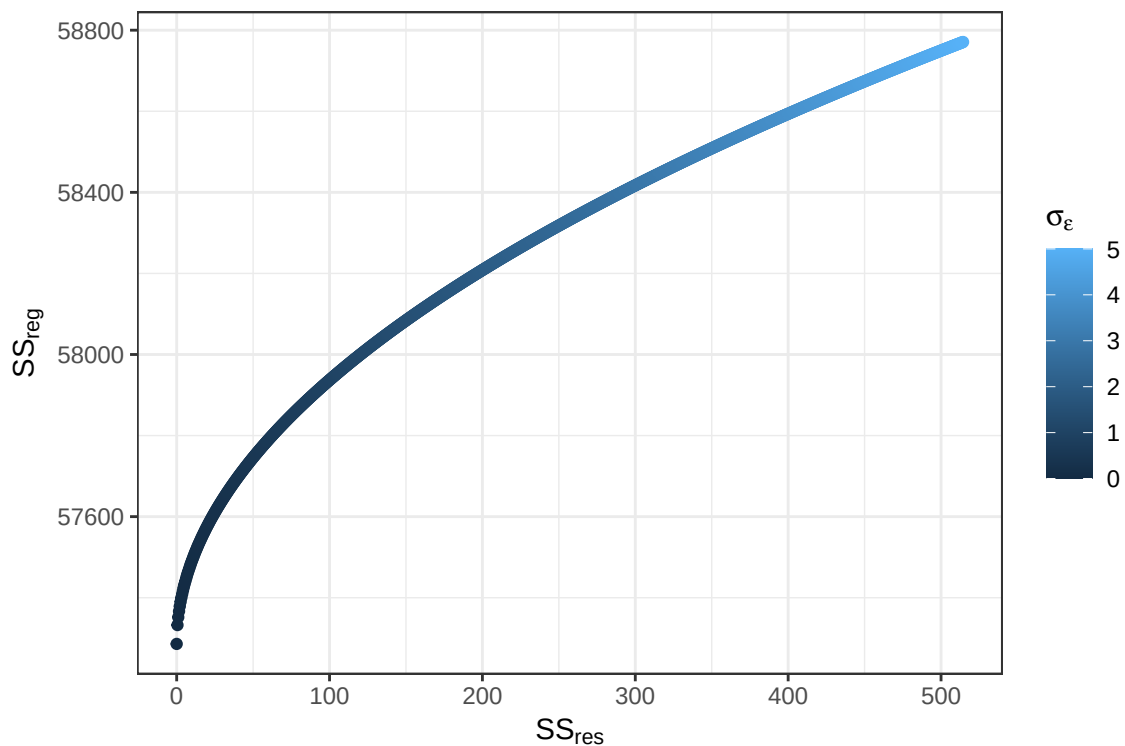


Figure 1: σ across levels of SS_{res} and SS_{reg} (fixed $\mathbf{x}$ and $\mathbf{Y}$)

We see that higher values of SS_{res} and SS_{reg} are associated with higher values of σ_ϵ . This makes sense as σ_ϵ affects the variation around how the data fits the model, and higher variation in σ_ϵ will result in higher levels of variation in the regression residuals and against $\hat{Y}$.

- (b) Use the `simulate.reg()` function to simulate regression data using the default set of coefficients. Create a plot that shows the distribution of the β s. Make sure you set `set.seed` to F so you can compare the output for differing samples of $\mathbf{x}$ and $\mathbf{Y}$. **What do you notice about their distributions?**

```

bsim <- tibble(Beta0 = rep(NA, 1000),
              Beta1 = rep(NA, 1000),
              Beta2 = rep(NA, 1000),
              Beta3 = rep(NA, 1000))

for (i in 1:1000){

  dat <- simulate.reg(set.seed=F)

  mod <- data.frame(summary(lm(y ~ x1 + x2 + x1*x2, data = dat))$coefficients)

  bsim$Beta0[i] <- mod$Estimate[1]
  bsim$Beta1[i] <- mod$Estimate[2]
  bsim$Beta2[i] <- mod$Estimate[3]
  bsim$Beta3[i] <- mod$Estimate[4]

}

bsim <- bsim |>
  pivot_longer(cols = colnames(bsim),
              names_to = "Beta",
              values_to = "Value")

library(ggribes)
ggplot(data = bsim, aes(x = Value, y = fct_rev(as.factor(Beta)))) +
  geom_density_ridges(alpha = 0.5) +
  theme_bw() +
  xlab("Estimate") +
  geom_hline(yintercept = 0) +
  ylab('') +
  scale_y_discrete(
labels = c(
  "Beta0" = bquote(beta[0]),
  "Beta1" = bquote(beta[1]),
  "Beta2" = bquote(beta[2]),
  "Beta3" = bquote(beta[3])
)
)

```

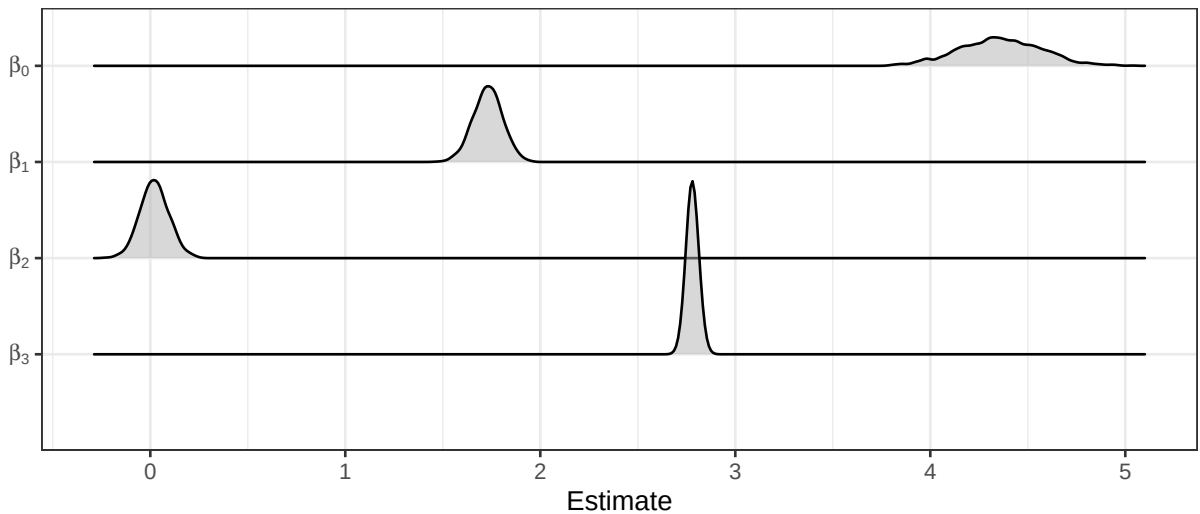


Figure 2: Distribution of the β coefficients

The spread of their distributions is very different. β_0 has the widest spread, followed by non-interaction terms β_1 and β_2 . Then, the coefficient with the least spread is clearly the interaction term β_3

- (c) Use the `simulate.reg()` function to simulate regression data using the default set of coefficients. Create a plot that shows the distribution of the $\hat{y}$ s at $x_1 = 3.6$ and $x_2 = -1.9$. Make sure you set

`set.seed` to F so you can compare the output for differing samples of **x** and **Y**. **What do you notice about the distribution?**

```

csim <- tibble(yhat = rep(NA, 1000))
for (i in 1:1000){

  dat <- simulate.reg(set.seed=F)

  mod <- data.frame(summary(lm(y ~ x1 + x2 + x1*x2, data = dat))$coefficients)

  betas <- mod$Estimate

  csim$yhat[i] <- betas[1] + 3.6*betas[2] - 1.9*betas[3] - 3.6*1.9*betas[4]

}

hist <- ggplot(data = csim, aes(x = yhat, y = after_stat(density))) +
  geom_histogram(bins = 15, color = "black", fill = "lightgrey") +
  theme_bw() +
  geom_hline(yintercept = 0) +
  xlab(bquote(hat(y))) +
  ylab('Density')
  scale_x_continuous(breaks = c(-10, -9, round(median(csim$yhat),2), -8, -7), limits = c(-10,-7))

## <ScaleContinuousPosition>
## Range:
## Limits: -10 -- -7

boxplot <- ggplot(data = csim, aes(x = "", y = yhat)) +
  geom_boxplot() +
  coord_flip() +
  xlab('') +
  ylab(bquote(hat(y))) +
  theme_bw() +
  scale_y_continuous(breaks = c(-10, -9, round(median(csim$yhat), 2), -8, -7), limits = c(-10,-7))

library(patchwork)
hist + boxplot +
  plot_layout(nrow = 2, heights = c(3, 1))

```

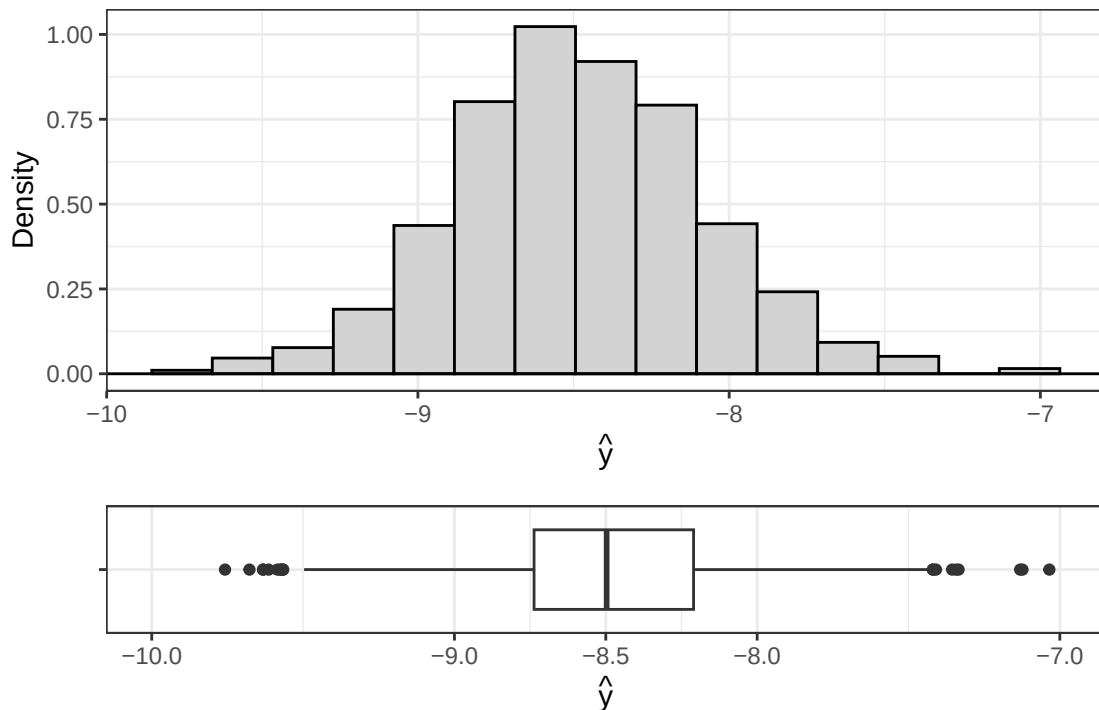


Figure 3: Distribution of $\hat{y}$'s at $x_1 = 3.6$ and $x_2 = -1.9$

The distribution of the $\hat{y}$ is Gaussian and centered at -8.5. The IQR is 0.53 [-8.74, -8.21] which is

similar to the initial parameter σ .

- (d) Describe why using t -based inference makes sense when conducting hypothesis tests or computing confidence intervals for a coefficient β_j , and the conditional expectation or prediction of Y ?

Based on the condition of linear regression that the residual ϵ 's are normally distributed, the β 's are calculated off of fixed data and random gaussian ϵ 's, so their distribution should be normal and therefore appropriate to t -test. Also based off of Figure: 1b, we can see that each of the β 's are normally distributed in a simulated context. The same is true for $\hat{y}$ in the way it's based off of fixed data and a normally distributed ϵ , and is clearly normally distributed in Figure: 1c.

2. Su et al. (2025) investigate how well-being of preschool teachers influences the effort they put into managing their emotions at work (emotional labor). They found that how committed a teacher is to their career (career commitment) plays a key role in linking well-being to emotional labor, and this connection gets stronger when they have more social support from colleagues or the community.

As part of this work, they aimed to replicate findings that the level of social support moderates the impact of preschool teachers' career commitment on emotional labor. Below, we will assess this piece to the puzzle.

Fit a regression model predicting **Emotional Labor** using **career commitment**, **social support**, and their interaction. Also include Gender, Age, and teaching experience.

```
dat <- read_csv('sudata-cleaned.csv')  
mod <- lm(EmotionalLabor ~ CareerCommitment*SocialSupport + Gender + Age + TeachingExp, data = dat)
```

- (a) Perform a residual analysis. Do you have any concerns about the model fit?

```
source('plotResiduals.R')
source('plotInfluence.R')

plotResiduals(mod)
```

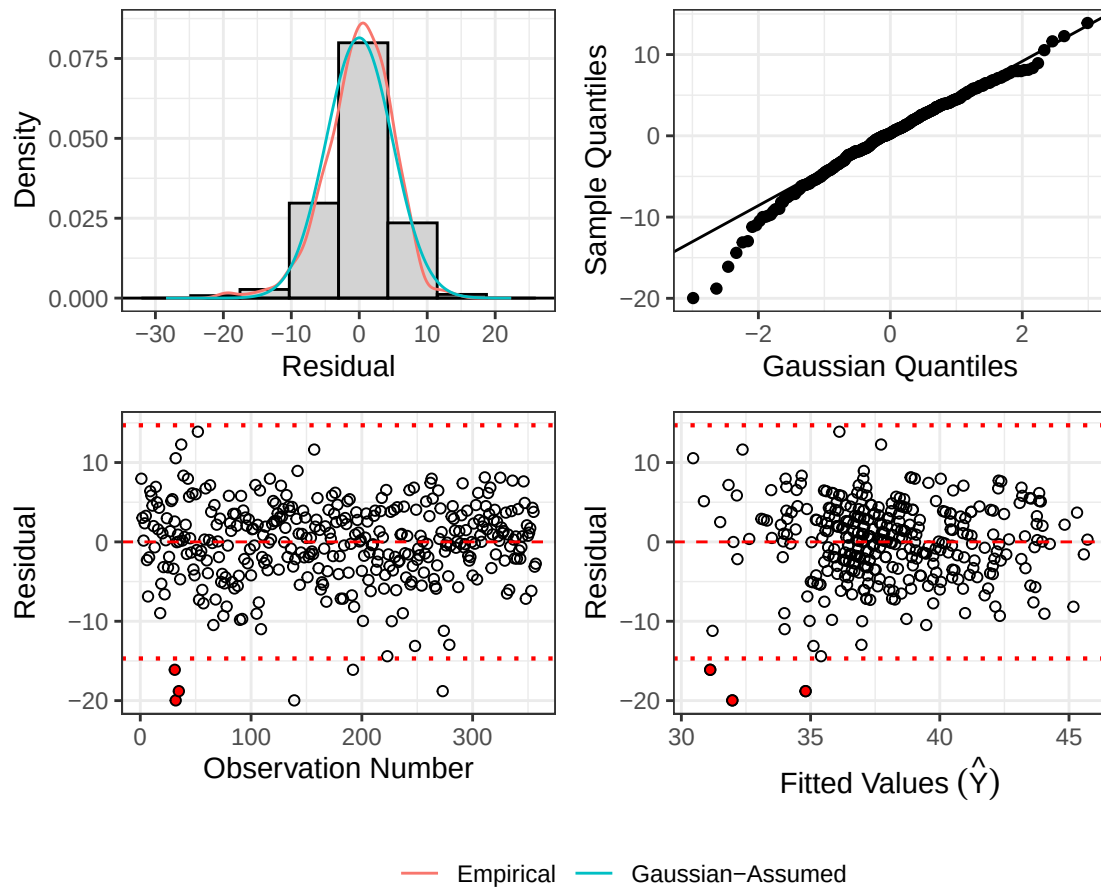


Figure 4: Residual Analysis of the Model

The residuals fit a normal distribution well as indicated by the histogram and the qqplot. There are 3 residuals that are potentially unusually low, but otherwise the residuals are scattered irregularly as we'd hope. I'd say this is very typical for a regression model of this size.

- (b) Perform an analysis of leverage/influence. Do you have any concerns about the model fit?

```
plotInfluence(mod)
```

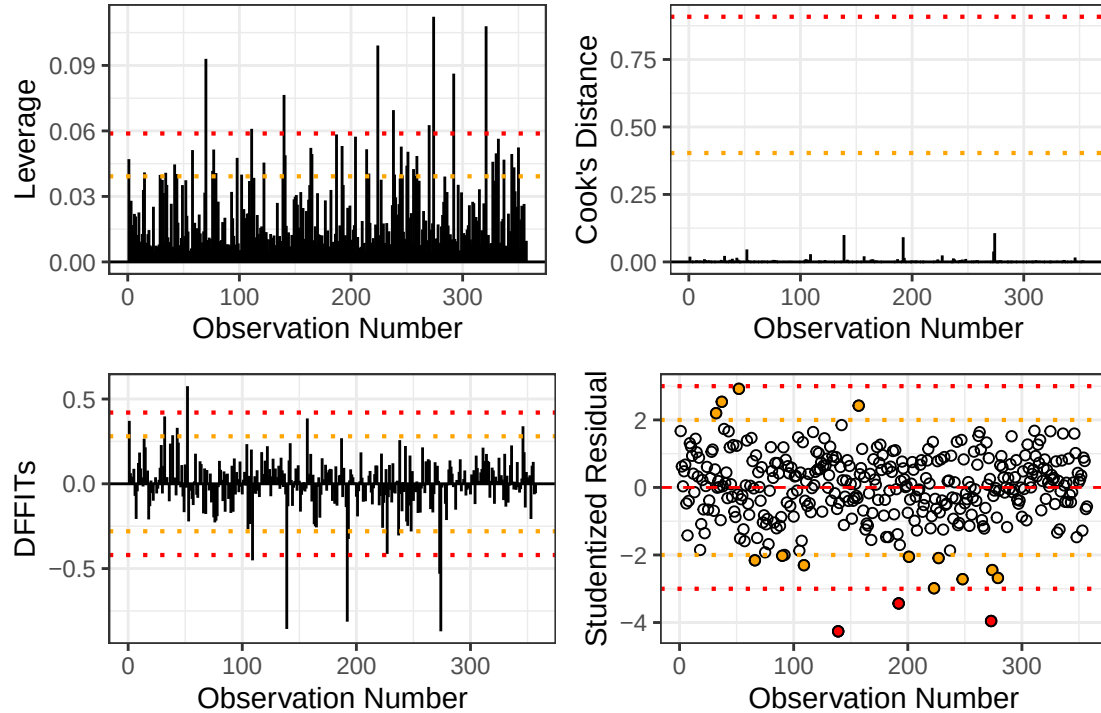


Figure 5: Leverage/Influence analysis of the model

There are several things to potentially be wary of here. There are 6-7 points with potentially high leverage, 4 points with large positive/negative DFFITS, and 3 points with large negative studentized residuals. For a positive, there are no points with large Cook's D. I'd maybe suggest testing a robust model to make sure these points don't heavily influence the model, but these results aren't super unusual for a model of this size.

- (c) Provide the results of the overall F test. What does this tell you about the model you fit?

```
library(xtable)
library(effectsize)

fctest <- summary(aov(mod))[[1]][-7,] |>
  mutate(significance = case_when('Pr(>F)' < 0.001 ~ "****",
    'Pr(>F)' >= 0.001 & 'Pr(>F)' < 0.01 ~ "**",
    'Pr(>F)' >= 0.01 & 'Pr(>F)' < 0.05 ~ "*",
    'Pr(>F)' >= 0.05 ~ ""))

omega2 = omega_squared(mod)$Omega2_partial,
interpretation = interpret_omega_squared(omega2, rules = "cohen1992"))

colnames(fctest) <- c("df", "Sum Sq", "Mean Sq", "F", "p-value",
  "Significance", "Effect Size", "Interpretation")
rownames(fctest) <- c("Career Commitment", "Social Support", "Gender", "Age", "Teaching Experience",
  "Career Commitment : Social Support")

fctest$p-value <- ifelse(fctest$p-value < 0.001, '<0.001', round(fctest$p-value, 3))

fctest$Effect Size <- ifelse(fctest$Effect Size == 0, '<0.001', round(fctest$Effect Size, 3))

fctest <- xtable(fctest, label = "model.f.tab", caption = "Model F-Test", digits = 3)
```

	df	Sum Sq	Mean Sq	F	p-value	Significance	Effect Size	Interpretation
Career Commitment	1.000	2657.030	2657.030	110.811	<0.001	***	0.235	medium
Social Support	1.000	3.910	3.910	0.163	0.687		<0.001	very small
Gender	1.000	23.192	23.192	0.967	0.326		<0.001	very small
Age	1.000	1.290	1.290	0.054	0.817		<0.001	very small
Teaching Experience	1.000	91.652	91.652	3.822	0.051		0.008	very small
Career Commitment : Social Support	1.000	186.770	186.770	7.789	0.006	**	0.019	very small

Table 1: Model F-Test

From the model fit, we see two terms that have a statistically discernible effect on Emotional Labor: Career Commitment ($F = 110.811$, $p < 0.001$, $\omega^2 = 0.235$) and the interaction between Career Commitment & Social Support ($F = 7.789$, $p = 0.006$, $\omega^2 = 0.019$). We can clearly see that Career Commitment has a direct effect on Emotional Labor, but we'll need further probing to investigate what's going on with the interaction term.

- (d) Based on the model fit, does the amount of social support impact the effect of career commitment on emotional labor?

```
#Low Social Support
coef.low <- data.frame(t(coef(mod))) |> # A data frame of coefficients
mutate(X.Intercept. = X.Intercept. +
  (mean(dat$SocialSupport)-sd(dat$SocialSupport)) * SocialSupport, # Adjust intercept and slope
  CareerCommitment = CareerCommitment +
  (mean(dat$SocialSupport)-sd(dat$SocialSupport)) * CareerCommitment.SocialSupport) |>
dplyr::select(-c(SocialSupport,CareerCommitment.SocialSupport))

# When Social Support is average
coef.avg <- data.frame(t(coef(mod))) |> # A data frame of coefficients
mutate(X.Intercept. = X.Intercept. +
  (mean(dat$SocialSupport)) * SocialSupport, # Adjust intercept and slope
  CareerCommitment = CareerCommitment +
  (mean(dat$SocialSupport)) * CareerCommitment.SocialSupport)|>
dplyr::select(-c(SocialSupport,CareerCommitment.SocialSupport))

# High Social Support
coef.high <- data.frame(t(coef(mod))) |> # A data frame of coefficients
mutate(X.Intercept. = X.Intercept. +
  (mean(dat$SocialSupport)+sd(dat$SocialSupport)) * SocialSupport, # Adjust intercept and slope
  CareerCommitment = CareerCommitment +
  (mean(dat$SocialSupport)+sd(dat$SocialSupport)) * CareerCommitment.SocialSupport)|>
dplyr::select(-c(SocialSupport,CareerCommitment.SocialSupport))

interaction_tab <- tibble('Social Support Level' = c("low (-1 SD)", "average (mean)", "high (+1 SD)"),
  'Career Commitment Slope Estimate' = c(coef.low$CareerCommitment,
    coef.avg$CareerCommitment,
    coef.high$CareerCommitment))

interaction_tab <- xtable(interaction_tab,label = "interaction.tab", caption = "Career Commitment and Social Support Interaction Summary")
```

Social Support Level	Career Commitment Slope Estimate
low (-1 SD)	0.284
average (mean)	0.396
high (+1 SD)	0.509

Table 2: Career Commitment and Social Support Interaction Summary (Averaged Across Gender)

We can see that the slope estimate of career commitment on emotional labor increases substantially as social support level increases.

3. Continue with the regression from Question 2.

- (a) Use `emmeans()` from the `emmeans` package for R (Lenth, 2024) to compute the estimated marginal means of Emotional Labor for low, average and high career commitment and social support. Report the results in a table and a figure.

```

library(emmeans)

emm <- data.frame(summary(emmeans(mod, ~ CareerCommitment*SocialSupport,
  at = list(
    CareerCommitment = c(mean(dat$CareerCommitment) - sd(dat$CareerCommitment),
                          mean(dat$CareerCommitment),
                          mean(dat$CareerCommitment) + sd(dat$CareerCommitment)),
    SocialSupport = c(mean(dat$SocialSupport) - sd(dat$SocialSupport),
                      mean(dat$SocialSupport),
                      mean(dat$SocialSupport) + sd(dat$SocialSupport))
  )))

emm_dat <- tibble(Career_Commitment = rep(c("Low", "Average", "High"), times = 3),
  Social_Support = rep(c("Low", "Average", "High"), each = 3),
  emmean = emm$emmean,
  CI_low = emm$lower.CL,
  CI_high = emm$upper.CL) |>
  mutate(Career_Commitment = factor(Career_Commitment, levels = c("Low", "Average", "High")),
  Social_Support = factor(Social_Support, levels = c("Low", "Average", "High")))

colnames(emm) <- c("Career Commitment", "Social Support",
  "EMM of Emotional Labor", "SE",
  "df", "Lower CL", "Upper CL")

emm <- xtable(emm, label = "emm.tab", caption = "Estimated Marginal Means (Averaged Across Gender)", digits = 3)

```

Career Commitment	Social Support	EMM of Emotional Labor	SE	df	Lower CL	Upper CL
17.445	31.429	35.565	0.585	350.000	34.414	36.716
24.036	31.429	37.434	0.549	350.000	36.355	38.514
30.626	31.429	39.304	0.767	350.000	37.796	40.812
17.445	38.934	34.955	0.566	350.000	33.842	36.068
24.036	38.934	37.566	0.501	350.000	36.580	38.552
30.626	38.934	40.177	0.583	350.000	39.029	41.324
17.445	46.438	34.345	0.766	350.000	32.838	35.852
24.036	46.438	37.697	0.613	350.000	36.492	38.902
30.626	46.438	41.049	0.659	350.000	39.753	42.345

Table 3: Estimated Marginal Means (Averaged Across Gender)

```
ggplot(data = emm_dat) +
  geom_point(aes(x = Social_Support,
                 y = emmean,
                 color = Career_Commitment),
            position = position_dodge(width = 0.5)) +
  geom_errorbar(aes(x = Social_Support,
                   ymin=CI_low,
                   ymax = CI_high,
                   color = Career_Commitment),
               position = position_dodge(width = 0.5),
               width = 0.1) +
  theme_bw() +
  labs(x = "Social Support",
       y = "Estimated Marginal Mean of Emotional Labor",
       color = "Career Commitment")
```

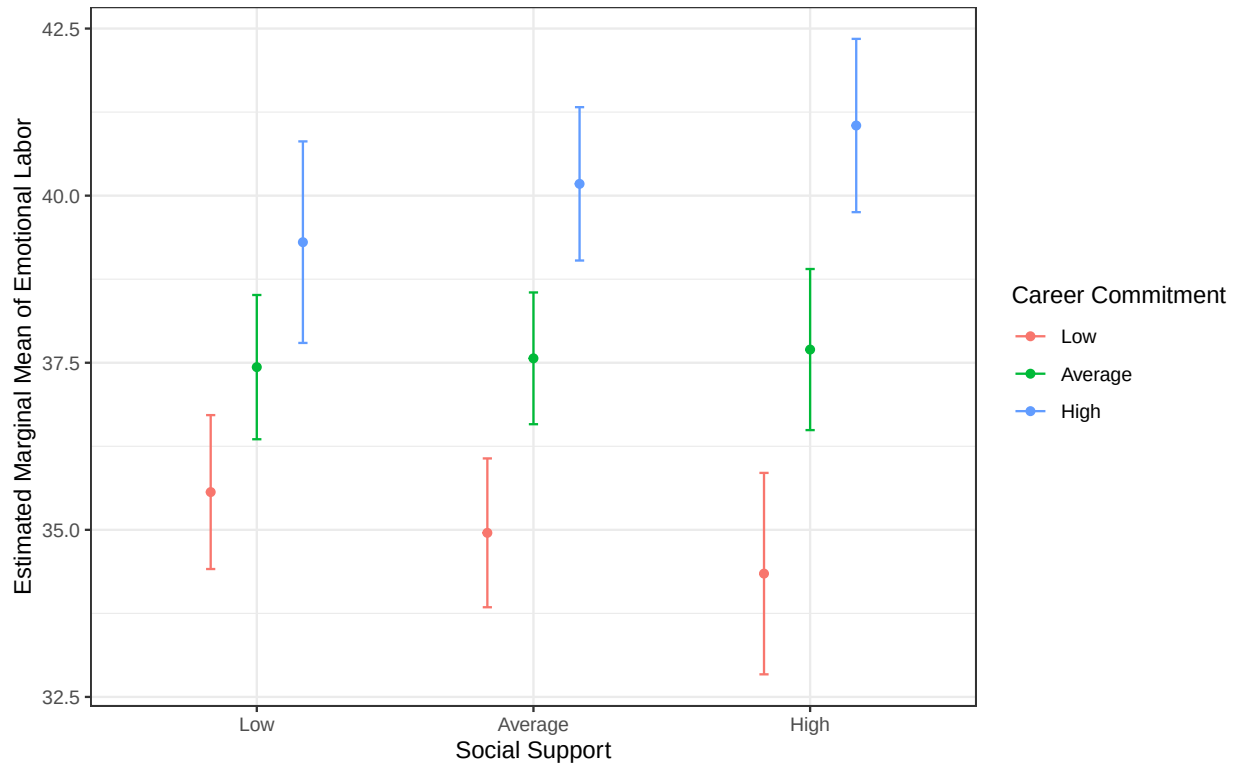


Figure 6: Estimated Marginal Means of Emotional Labor Across Low (Mean-SD), Average (Mean), and High (Mean+SD) Levels of Career Commitment and Social Support (Averaged Across Gender)

- (b) Use `emtrends()` from the `emmeans` package for R (Lenth, 2024) to compute the slope for **career commitment** at low, average, and high **social support**. Report the results in a table and use `ggeffect()` from the `ggeffects` package for R (Lüdtke, 2018) to create a plot.

```
emt <- data.frame(test(emtrends(mod, ~ CareerCommitment*SocialSupport, var="CareerCommitment",
                             at = list(SocialSupport = c(mean(dat$SocialSupport) - sd(dat$SocialSupport),
                                                         mean(dat$SocialSupport),
                                                         mean(dat$SocialSupport) + sd(dat$SocialSupport))))))

colnames(emt) <- c("Career Commitment", "Social Support",
                  "Career Commitment Slope", "SE",
                  "df", "t", "p-value")

emt$p-value <- ifelse(emt$p-value < 0.001, "< 0.001", emt$p-value)

emt <- xtable(emt, label = "emt.tab", caption = "Estimated Marginal Slope of Career Commitment
Across Low (Mean-SD), Average (Mean), and High (Mean+SD)
Levels of Social Support (Averaged
```

Across Gender)", digits = 3)

Career Commitment	Social Support	Career Commitment Slope	SE	df	t	p-value
23.994	31.429	0.284	0.061	350.000	4.614	< 0.001
23.994	38.934	0.396	0.043	350.000	9.278	< 0.001
23.994	46.438	0.509	0.056	350.000	9.114	< 0.001

Table 4: Estimated Marginal Slope of Career Commitment Across Low (Mean-SD), Average (Mean), and High (Mean+SD) Levels of Social Support (Averaged Across Gender)

```
library(ggeffects)

ggeff <- ggeffect(mod, ~ CareerCommitment*SocialSupport,
  at = list(SocialSupport = c(mean(dat$SocialSupport) - sd(dat$SocialSupport),
    mean(dat$SocialSupport),
    mean(dat$SocialSupport) + sd(dat$SocialSupport)))) |>
  tibble()

ggplot(data=ggeff)+
  geom_line(aes(x=x, y=predicted, color=group))+
  geom_ribbon(aes(x=x, ymin=conf.low, ymax=conf.high, fill=group),
    alpha=0.1)+
  theme_bw()+
  ylab("Predicted Emotional Labor") +
  xlab("Career Commitment") +
  scale_color_discrete("Social Support", labels = c("Low", "Average", "High")) +
  guides(fill = "none") +
  labs(color = "Social Support")
```

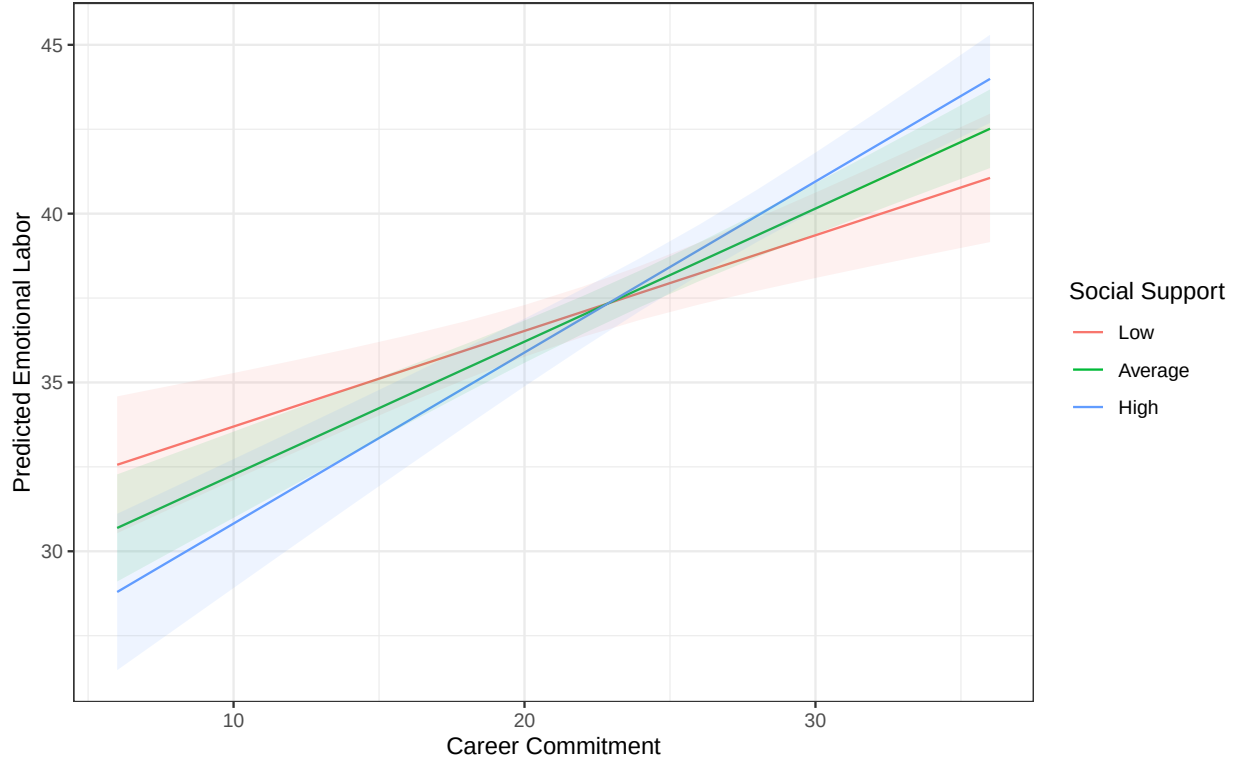


Figure 7: Marginal Trend of Career Commitment across Low(Mean-SD), Average(Mean), and High(Mean+SD) Levels of Social Support (Averaged Across Gender)

- (c) Use `pairs()` to compare the slopes for `career commitment` at low, average, and high social support. What does this tell us?

```
pairs <- data.frame(summary(pairs(emtrends(mod, ~ CareerCommitment*SocialSupport, var="CareerCommitment",
  at = list(SocialSupport = c(mean(dat$SocialSupport)-sd(dat$SocialSupport),
    mean(dat$SocialSupport),
    mean(dat$SocialSupport) + sd(dat$SocialSupport)))))))

pairs$contrast <- c("Low - Average", "Low - High", "Average - High")

colnames(pairs) <- c("Contrast (Social Support)", "Slope Difference",
  "SE", "df",
  "t", "p-value")

pairs <- xtable(pairs, label = "pairs.tab", caption = "Contrasts of Career Commitment Slopes at
  Low(Mean-SD), Average(Mean),
  and High(Mean+SD) Levels of Social Support (Averaged Across Gender)", digits = 3)
```

Contrast (Social Support)	Slope Difference	SE	df	t	p-value
Low - Average	-0.112	0.040	350.000	-2.791	0.015
Low - High	-0.225	0.081	350.000	-2.791	0.015
Average - High	-0.112	0.040	350.000	-2.791	0.015

Table 5: Contrasts of Career Commitment Slopes at Low(Mean-SD), Average(Mean), and High(Mean+SD) Levels of Social Support (Averaged Across Gender)

Each of the pairwise comparisons of Career Commitment slopes is discernibly different, indicating that the effect of Social Support on Career Commitment generally extends far across every of Social Support. There isn't, for example, a point where the effect of Social Support on Career Commitment doesn't seem to make much of a difference anymore, the effect is pretty linear across the whole model.

- (d) Use `johnson_neyman()` from the `interactions` package for R (Long, 2019) to evaluate which values of social support yield slopes for `career commitment` that are statistically discernible from zero. What does this tell us?

```
library(interactions)
johnson_neyman(mod,
               pred="CareerCommitment", modx="SocialSupport",
               control.fdr = T,
               modx.label = "Social Support",
               y.label = "Slope of Career Commitment")

## JOHNSON-NEYMAN INTERVAL
##
## When SocialSupport is OUTSIDE the interval [-63.81, 24.86], the slope of
## CareerCommitment is p < .05.
##
## Note: The range of observed values of SocialSupport is [18.00, 56.00]
##
## Interval calculated using false discovery rate adjusted t = 2.05
```

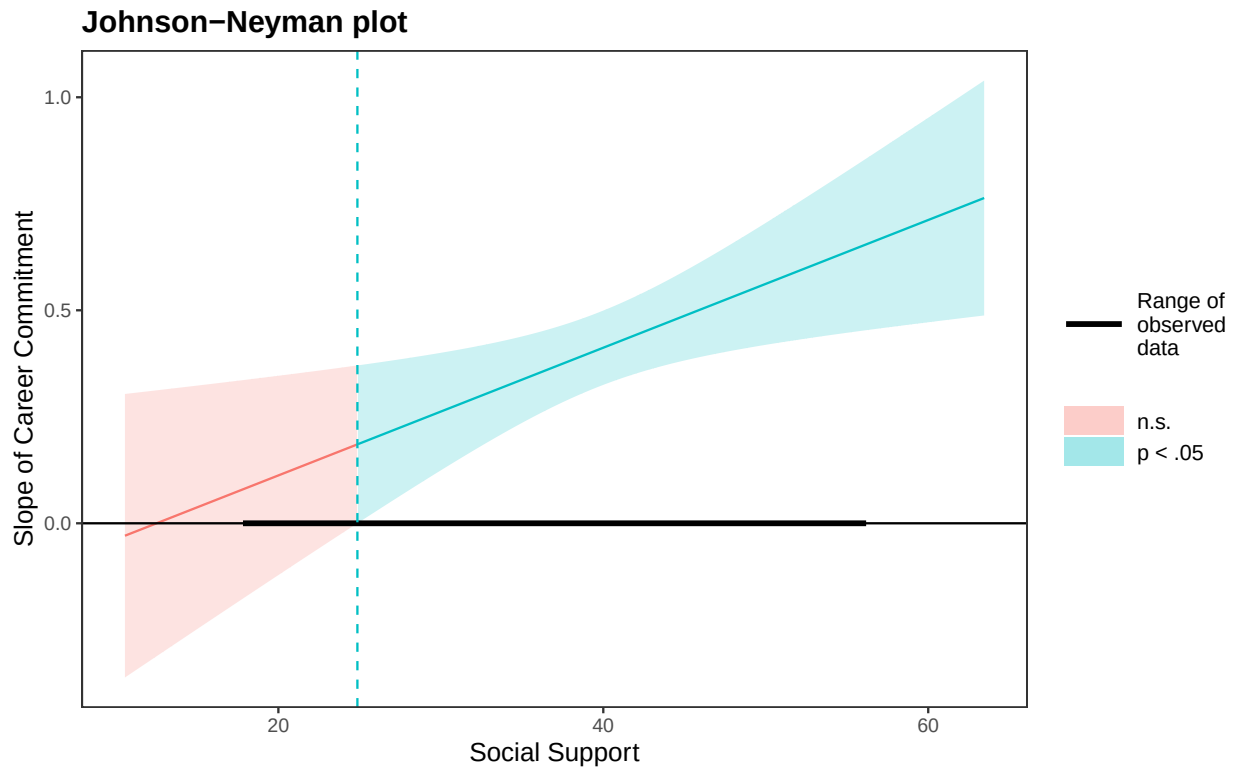


Figure 8: Johnson-Neyman Plot of Statistical Discernibility of the Slope of Career Commitment Across Values of Social Support (Averaged Across Gender)

From the Johnson-Neyman plot, we can see that in the range of our observed data, the effect of Career Commitment on Emotional Labor is significant for levels of Social Support > 24.86. The slope of Career Commitment is linearly monotone increasing as well, which suggests that the effect of Career Commitment on Emotional Labor is heavily impacted by having a high level of Social Support and for people in the sample with low levels of Social Support, their Career Commitment doesn't discernibly affect their Emotional Labor.

References

- Lenth, R. V. (2024). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. R package version 1.10.3.
- Long, J. A. (2019). *interactions: Comprehensive, User-Friendly Toolkit for Probing Interactions*. R package version 1.1.0.

- Lüdtke, D. (2018). ggeffects: Tidy data frames of marginal effects from regression models. *Journal of Open Source Software*, 3(26):772.
- Su, X., Yu, D., Yu, Y., and Lian, R. (2025). Well-being and emotional labor for preschool teachers: The mediation of career commitment and the moderation of social support. *PLoS One*, 20(4):e0318027.